

Program 1 : Database Connection Using JDBC in Java

```
import javax.swing.*.*;

import java.awt.event.*;

import java.sql.*;

public class RegisterForm {

    public static void main(String[] args) {

        JFrame frame = new JFrame("Student Registration Form");

        // Name
        JLabel label = new JLabel("Enter Full Name : ");
        label.setBounds(50, 50, 150, 20);

        JTextField textField = new JTextField();
        textField.setBounds(200, 50, 150, 20);

        // Year
        JLabel label2 = new JLabel("Select Year : ");
        label2.setBounds(50, 80, 150, 20);

        JRadioButton radioButton1 = new JRadioButton("FY");
        radioButton1.setBounds(200, 80, 50, 20);

        JRadioButton radioButton2 = new JRadioButton("SY");
        radioButton2.setBounds(260, 80, 50, 20);

        JRadioButton radioButton3 = new JRadioButton("TY");
        radioButton3.setBounds(320, 80, 50, 20);
```

```
JRadioButton radioButton4 = new JRadioButton("BTech");  
radioButton4.setBounds(380, 80, 80, 20);
```

```
ButtonGroup radioGroup = new ButtonGroup();  
radioGroup.add(radioButton1);  
radioGroup.add(radioButton2);  
radioGroup.add(radioButton3);  
radioGroup.add(radioButton4);
```

```
// Subjects
```

```
JLabel label3 = new JLabel("Select Subjects : ");  
label3.setBounds(50, 110, 150, 20);
```

```
JCheckBox checkBox1 = new JCheckBox("Java");  
checkBox1.setBounds(200, 110, 70, 20);
```

```
JCheckBox checkBox2 = new JCheckBox("DSA");  
checkBox2.setBounds(270, 110, 70, 20);
```

```
JCheckBox checkBox3 = new JCheckBox("DBMS");  
checkBox3.setBounds(340, 110, 80, 20);
```

```
// Buttons
```

```
JButton submitButton = new JButton("Submit");  
submitButton.setBounds(200, 150, 100, 30);
```

```
JButton clearButton = new JButton("Clear");  
clearButton.setBounds(320, 150, 100, 30);
```

```
// Add components
```

```
frame.add(label);
```

```
frame.add(textField);
frame.add(label2);
frame.add(radioButton1);
frame.add(radioButton2);
frame.add(radioButton3);
frame.add(radioButton4);
frame.add(label3);
frame.add(checkBox1);
frame.add(checkBox2);
frame.add(checkBox3);
frame.add(submitButton);
frame.add(clearButton);
```

```
// Submit Button Logic (MySQL Insert)
```

```
submitButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
```

```
        String name = textField.getText();
```

```
        String year = "";
```

```
        if (radioButton1.isSelected()) year = "FY";
```

```
        else if (radioButton2.isSelected()) year = "SY";
```

```
        else if (radioButton3.isSelected()) year = "TY";
```

```
        else if (radioButton4.isSelected()) year = "BTech";
```

```
        String subjects = "";
```

```
        if (checkBox1.isSelected()) subjects += "Java ";
```

```
        if (checkBox2.isSelected()) subjects += "DSA ";
```

```
        if (checkBox3.isSelected()) subjects += "DBMS ";
```

```
        try {
```

```
// Load Driver
Class.forName("com.mysql.cj.jdbc.Driver");

// Connect DB
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/college", "root", "root"
);

// SQL Insert
PreparedStatement ps = con.prepareStatement(
    "INSERT INTO students(name, year, subjects) VALUES (?, ?, ?)"
);

ps.setString(1, name);
ps.setString(2, year);
ps.setString(3, subjects);

ps.executeUpdate();

JOptionPane.showMessageDialog(frame, "Data Inserted Successfully!");

con.close();

} catch (Exception ex) {
    ex.printStackTrace();
    JOptionPane.showMessageDialog(frame, "Error: " + ex.getMessage());
}
}

});

// Clear Button Logic
```

```

clearButton.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        textField.setText("");

        radioGroup.clearSelection();

        checkBox1.setSelected(false);

        checkBox2.setSelected(false);

        checkBox3.setSelected(false);

    }

});

// Frame settings

frame.setSize(500, 400);

frame.setLayout(null);

frame.setVisible(true);

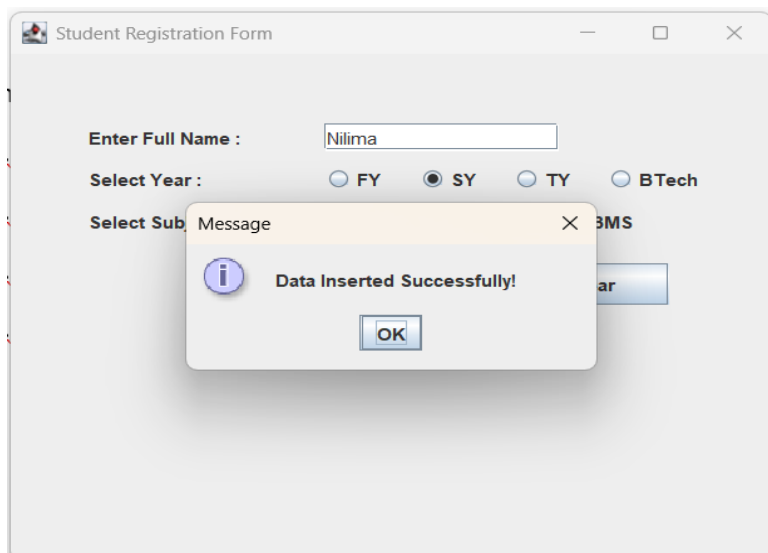
frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

}

}

```

Output :



Database Creation :

CREATE DATABASE college;

USE college;

CREATE TABLE students (

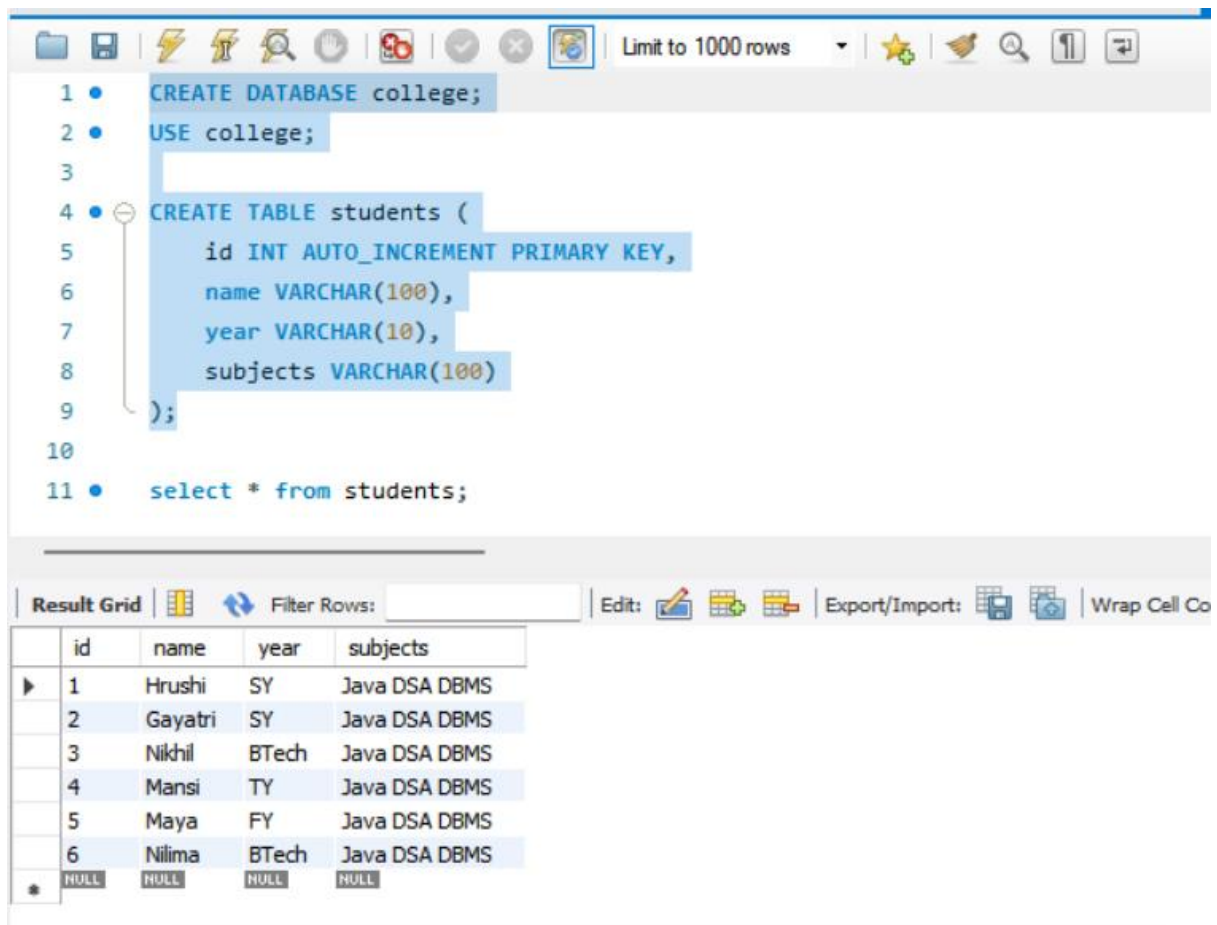
id INT AUTO_INCREMENT PRIMARY KEY,

name VARCHAR(100),

year VARCHAR(10),

subjects VARCHAR(100)

);



The screenshot shows a database management tool interface. The top toolbar includes icons for file operations, execution, and search. The main area displays a list of SQL queries with line numbers 1 through 11. The queries are:

```
1 • CREATE DATABASE college;
2 • USE college;
3
4 • CREATE TABLE students (
5     id INT AUTO_INCREMENT PRIMARY KEY,
6     name VARCHAR(100),
7     year VARCHAR(10),
8     subjects VARCHAR(100)
9 );
10
11 • select * from students;
```

Below the queries, the 'Result Grid' is visible, showing the output of the last query. It includes a toolbar with 'Filter Rows', 'Edit', 'Export/Import', and 'Wrap Cell Co'. The result grid contains the following data:

	id	name	year	subjects
▶	1	Hrushu	SY	Java DSA DBMS
	2	Gayatri	SY	Java DSA DBMS
	3	Nikhil	BTech	Java DSA DBMS
	4	Mansi	TY	Java DSA DBMS
	5	Maya	FY	Java DSA DBMS
	6	Nilima	BTech	Java DSA DBMS
*	NULL	NULL	NULL	NULL